

COMPANION FOR CHAPTER 3

OF

**FUNDAMENTALS
OF PROBABILITY**

WITH STOCHASTIC PROCESSES

FOURTH EDITION

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Chapter 3

Conditional Probability and Independence

3A ADDITIONAL EXAMPLES

Example 3a What is the probability that among n unrelated people, selected at random, no two have the same birthday? Assume that the birth rates are constant throughout the year and that each year has 365 days.

Solution: The probability that n unrelated randomly selected people have n different birthdays is

$$\left(1 - \frac{1}{365}\right)\left(1 - \frac{2}{365}\right)\left(1 - \frac{3}{365}\right) \cdots \left(1 - \frac{n-1}{365}\right). \quad \blacklozenge$$

Example 3b At a party, there are three coolers. The first one has 10 regular colas, 12 diet colas, 5 orange drinks, and 7 ginger ales. The second one has 13 diet colas and 12 ginger ales. The third cooler has 15 regular colas, 5 ginger ales, and 8 orange drinks. Mali chooses a cooler at random and draws a drink from the cooler without looking. What is the probability that she draws a diet cola?

Solution: For $1 \leq i \leq 3$, let A_i be the event that Mali chooses the i th cooler. Let D be the event that she draws a diet cola. The desired probability is

$$\begin{aligned} P(D) &= P(D \mid A_1)P(A_1) + P(D \mid A_2)P(A_2) + P(D \mid A_3)P(A_3) \\ &= \frac{12}{34} \cdot \frac{1}{3} + \frac{13}{25} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} = \frac{371}{1275} \approx 0.29. \quad \blacklozenge \end{aligned}$$

Example 3c Suppose that for infectious diseases, when a patient is tested for a specific disease, the result is not 100% reliable, and 7% of the time it is false negative. That is, a patient that has the disease is not actually diagnosed with it. However, if the patient does not have the disease, the test is 100% reliable. Fontana is experiencing common symptoms of three infectious diseases A , B , and C . Her doctor tested her for disease A , and the result was negative. If the chances are 60%, 27% and 13% that patients with those symptoms have disease A , B , and C , respectively, what is the probability that Fontana has disease A ? Assume that no one gets infected with more than one of these infectious diseases.

Solution: Let E , F and G be the events that Fontana has disease A , B , and C , respectively. Let H be the event that she has tested negative for disease A . The desired probability is

$$\begin{aligned} P(E | H) &= \frac{P(H | E)P(E)}{P(H | E)P(E) + P(H | F)P(F) + P(H | G)P(G)} \\ &= \frac{(0.07)(0.60)}{(0.07)(0.60) + 1 \cdot (0.27) + 1 \cdot (0.13)} = 0.095. \quad \blacklozenge \end{aligned}$$

Example 3d A number is selected at random from the set $\{1, 2, \dots, 20\}$. Then a second number is selected randomly between 1 and the first number. If the second number is 5, what is the probability the first number is 15?

Solution: For $1 \leq i \leq 20$, let A_i be the event that the first number selected is i . Let F be the event that the second number is 5. By Bayes' Theorem,

$$P(A_{15} | F) = \frac{P(F | A_{15})P(A_{15})}{\sum_{i=1}^{20} P(F | A_i)P(A_i)} = \frac{\frac{1}{15} \cdot \frac{1}{20}}{\sum_{i=5}^{20} \frac{1}{i} \cdot \frac{1}{20}} \approx 0.044. \quad \blacklozenge$$

Example 3e For its cars, an automobile assembly manufacturing plant purchases engines from two different auto engine manufactures: 35% from Wilbraham Motors and 65% from Antonio Engines. Even though the warranties offered by both companies are identical, experience shows that only 87% of the engines manufactured by Wilbraham Motors last at least for 150,000 miles whereas 96% of those manufactured by Antonio Engines last for that many miles. (a) Find the probability that the engine of a randomly selected car assembled by this manufacturing plant lasts at least 150,000 miles. (b) If the engine of a random car manufactured by this plant failed before functioning for 150,000 miles, what is the probability that it was manufactured by Wilbraham Motors?

Solution:

- (a) Let A be the event that the engine of the random car manufactured by this plant lasts at least 150,000 miles. Let B be the event that the engine of the randomly selected car was manufactured by Wilbraham Motors. We have

$$\begin{aligned} P(A) &= P(A | B)P(B) + P(A | B^c)P(B^c) \\ &= (0.87)(0.35) + (0.96)(0.65) \approx 0.93. \end{aligned}$$

(b)

$$\begin{aligned} P(B | A^c) &= \frac{P(A^c | B)P(B)}{P(A^c | B)P(B) + P(A^c | B^c)P(B^c)} \\ &= \frac{(1 - 0.87)(0.35)}{(1 - 0.87)(0.35) + (1 - 0.96)(0.65)} \approx 0.64. \quad \blacklozenge \end{aligned}$$

Example 3f Solve the following problem, asked of Marilyn Vos Savant in the “Ask Marilyn” column of *Parade Magazine*, January 3, 1999.

You’re at a party with 199 other guests when robbers break in and announce that they are going to rob one of you. They put 199 blank pieces of paper in a hat, plus one marked “you lose.” Each guest must draw, and the person who draws “you lose” will get robbed. The robbers offer you the option of drawing first, last, or at any time in between. When would you take your turn?

Assume that those who draw earlier will not announce whether they drew the “you lose” paper.

Solution: Let A_i be the event that the i th person draws the “you lose” paper. Clearly,

$$P(A_1) = \frac{1}{200},$$

$$P(A_2) = P(A_1^c A_2) = P(A_1^c)P(A_2 | A_1^c) = \frac{199}{200} \cdot \frac{1}{199} = \frac{1}{200},$$

$$P(A_3) = P(A_1^c A_2^c A_3) = P(A_1^c)P(A_2^c | A_1^c)P(A_3 | A_1^c A_2^c) = \frac{199}{200} \cdot \frac{198}{199} \cdot \frac{1}{198} = \frac{1}{200},$$

and so on. Therefore, $P(A_i) = 1/200$ for $1 \leq i \leq 200$. This means that it makes no difference if you draw first, last or anywhere in the middle. Here is Marilyn Vos Savant’s intuitive solution to this problem:

It makes no difference if you draw first, last, or anywhere in the middle. Look at it this way: Say the robbers make everyone draw at once. You’d agree that everyone has the same chance of losing (one in 200), right? Taking turns just makes that same event happen in a slow and orderly fashion. Envision a raffle at a church with 200 people in attendance, each person buys a ticket. Some buy a ticket when they arrive, some during the event, and some just before the winner is drawn. It doesn’t matter. At the party the end result is this: all 200 guests draw a slip of paper, and, regardless of when they look at the slips, the result will be identical: one will lose. You can’t alter your chances by looking at your slip before anyone else does, or waiting until everyone else has looked at theirs. ♦

Example 3g At a university, 30% of the students are freshmen, 25% are sophomores, 23% are juniors, and the remaining 22% are seniors. Suppose that 7% of the freshmen, 6% of the sophomores, 5% of the juniors, and 8% of the seniors are A-students. If a randomly selected student from this university is an A-student, what is the probability that he or she is a senior?

Solution: Let E , F , G , and H be the events that the randomly selected student is a freshman, sophomore, junior, and senior, respectively. Let L be the event that the student is an

A-student. We are interested in $P(H | L)$. By Bayes theorem,

$$\begin{aligned}
 P(H | L) &= \frac{P(L | H)P(H)}{P(L | E)P(E) + P(L | F)P(F) + P(L | G)P(G) + P(L | H)P(H)} \\
 &= \frac{(0.08)(0.22)}{(0.07)(0.30) + (0.06)(0.25) + (0.05)(0.23) + (0.08)(0.22)} = 0.27. \quad \blacklozenge
 \end{aligned}$$

Example 3h Suppose that 16% of an insurance company's automobile policyholders are male and under the age of 25, while 12% are female and under the age of 25. The following table lists the percentages of various groups of policyholders who were involved in a car accident last a year.

Group	Male Under 25	Female Under 25	Between 25 and 65	Over 65
Percentage of Accidents	20%	8%	5%	10%

Find the range of the percentages of this company's policyholders who got involved in a car accident during the previous year.

Solution: Let M , F , B , and O be, respectively, the events that a randomly selected automobile policyholder of this company is a male and under the age of 25, a female and under the age of 25, between the ages 25 and 65, and over the age of 65. Clearly, $\{M, F, B, O\}$ is a partition of the sample space. Thus,

$$P(M) + P(F) + P(B) + P(O) = 1,$$

implying that

$$0.16 + 0.12 + P(B) + P(O) = 1.$$

This gives $P(B) + P(O) = 0.72$. Let A be the event that the randomly selected policyholder got involved in an accident last year. We want to find the greatest lower bound and the least upper bound for $P(A)$. By the law of total probability,

$$\begin{aligned}
 P(A) &= P(A | M)P(M) + P(A | F)P(F) + P(A | B)P(B) + P(A | O)P(O) \\
 &= (0.20)(0.16) + (0.08)(0.12) + (0.05)P(B) + (0.10)P(O) \\
 &= 0.0416 + (0.05)P(B) + (0.10)P(O).
 \end{aligned}$$

The greatest lower bound for $P(A)$ is obtained, from this relation, by substituting 0.72 for $P(B)$ and 0 for $P(O)$:

$$P(A) > 0.0416 + (0.05)(0.72) = 0.0776.$$

Similarly, the least upper bound for $P(A)$ is obtained, from the relation, by substituting 0 for $P(B)$ and 0.72 for $P(O)$:

$$P(A) < 0.0416 + (0.10)(0.72) = 0.1136.$$

Therefore, the range of the percentages we are interested in is between 7.76% and 11.36%. Note that the data in the table implies that neither $P(O)$ nor $P(B)$ can be 0. So the inequalities above are both strict. ♦

Example 3i Suppose that $n \geq 2$ missiles are fired at a target and hit it independently. If the probability that the i th missile hits it is p_i , $i = 1, 2, \dots, n$, find the probability that at least two missiles will hit the target.

Solution: For convenience, let $p_0 = 0$; the desired probability is

$$1 - P(\text{none of the missiles hits the target}) - P(\text{exactly one of the missiles hits the target}) \\ = 1 - \prod_{i=1}^n (1 - p_i) - \sum_{i=1}^n (1 - p_1)(1 - p_2) \cdots (1 - p_{i-1})p_i(1 - p_{i+1}) \cdots (1 - p_n). \quad \blacklozenge$$

Example 3j Avril has certain standards for selecting her future husband. She has n suitors and knows how to compare any two and rank them. She decides to date one suitor at a time randomly. When she knows a suitor well enough, she can marry or reject him. If she marries the suitor, she can never know the ones not dated. If she rejects the suitor, she will not be able to reconsider him. In this process, Avril will not choose a suitor she ranks lower than at least one of the previous ones dated. Avril's goal is to maximize the probability of selecting the best suitor. To achieve this, she adopts the following strategy: For some m , $0 \leq m < n$, she dumps the first m suitors she dates after knowing each of them well no matter how good they are. Then she marries the first suitor she dates who is better than all those preceding him. In terms of n , find the value of m which maximizes the probability of selecting the best suitor.

Remark: This example is a model for many real-world problems in which we have to reject “choices” or “offers” until the one we think is the “best.” Note that it is quite possible that we reject the best “offer” or “choice” in the process. If this happens, we will not be able to make a selection.

Solution: Let E_m be the event that Avril selects the best suitor given her strategy. Let B_i be the event that the best suitor is the i th of Avril's dates. By the law of total probability,

$$P(E_m) = \sum_{i=1}^n P(E_m | B_i)P(B_i) = \frac{1}{n} \sum_{i=1}^n P(E_m | B_i).$$

Clearly, $P(E_m | B_i) = 0$ for $1 \leq i \leq m$. For $i > m$, if the i th suitor is the best, then Avril chooses him if and only if among the first $i - 1$ suitors Avril dates, the best is one of the first m . So

$$P(E_m | B_i) = \frac{m}{i - 1}.$$

Therefore,

$$P(E_m) = \frac{1}{n} \sum_{i=m+1}^n \frac{m}{i - 1} = \frac{m}{n} \sum_{i=m+1}^n \frac{1}{i - 1}.$$

Now

$$\sum_{i=m+1}^n \frac{1}{i-1} \approx \int_m^n \frac{1}{x} dx = \ln\left(\frac{n}{m}\right).$$

Thus

$$P(E_m) \approx \frac{m}{n} \ln\left(\frac{n}{m}\right).$$

To find the maximum of $P(E_m)$, consider the differentiable function

$$h(x) = \frac{x}{n} \ln\left(\frac{n}{x}\right).$$

Since

$$h'(x) = \frac{1}{n} \ln\left(\frac{n}{x}\right) - \frac{1}{n} = 0$$

implies that $x = n/e$, the maximum of $P(E_m)$ is at $m = [n/e]$, where $[n/e]$ is the greatest integer less than or equal to n/e . Hence Avril should dump the first $[n/e]$ suitors she dates and marry the first suitor she dates afterward who is better than all those preceding him. The probability that with such a strategy she selects the best suitor of all n is approximately

$$h\left(\frac{n}{e}\right) = \frac{1}{e} \ln e = \frac{1}{e} \approx 0.368. \quad \blacklozenge$$

Example 3k (Shrewd Prisoner's Dilemma) Because of a prisoner's constant supplication, the king grants him this favor: He is given $2N$ balls, which differ from each other only in that half of them are green and half are red. The king instructs the prisoner to divide the balls between two identical urns. One of the urns will then be selected at random, and the prisoner will be asked to choose a ball at random from the urn chosen. If the ball turns out to be green, the prisoner will be freed. How should he distribute the balls in the urn to maximize his chances of freedom?

Hint: *Solution:* Let g be the number of green balls and r be the number of red balls

in the first urn. The corresponding numbers in the second urn are $N - g$ and $N - r$. The probability that a green ball is drawn is $f(g, r)$:

$$f(g, r) = \frac{1}{2} \left(\frac{g}{g+r} + \frac{N-g}{2N-g-r} \right).$$

We will find the maximum of this function of two variables (r and g). Note that the maximum need not occur at an interior point of the domain. Let $\mathbf{N}$ be the set of nonnegative integers. The domain of f is

$$\{(g, r) \in \mathbf{N} \times \mathbf{N} : 0 \leq g \leq N, 0 \leq r \leq N, 0 < g+r < 2N\}.$$

Extending the domain of f to all points $(g, r) \in \mathbf{R} \times \mathbf{R}$, we find that $\frac{\partial f}{\partial g} = \frac{\partial f}{\partial r} = 0$ gives $g = r = N/2$ and $f(N/2, N/2) = 1/2$. However, this is not the maximum value because on the boundary of the domain of f along $r = 0$, we find that

$$f(g, 0) = \frac{1}{2} \left(1 + \frac{N-g}{2N-g} \right)$$

is maximum at $g = 1$ and

$$f(1, 0) = \frac{1}{2} \left(\frac{3N - 2}{2N - 1} \right) \geq \frac{1}{2}.$$

We also find that on the boundary along $r = N$,

$$f(g, N) = \frac{1}{2} \left(\frac{g}{g + N} + 1 \right)$$

is maximum at $g = N - 1$ and

$$f(N - 1, N) = \frac{1}{2} \left(\frac{3N - 2}{2N - 1} \right) \geq \frac{1}{2}.$$

The maximums of f along other sides of the boundary are all less than $\frac{1}{2} \left(\frac{3N - 2}{2N - 1} \right)$. Therefore, there are exactly two maximums and they occur at $(1, 0)$ and $(N - 1, N)$. That is, the maximum of f occurs if one urn contains one green and 0 red balls and the other one contains $N - 1$ green and N red balls. For large N , the probability that the prisoner is freed is $\frac{1}{2} \left(\frac{3N - 2}{2N - 1} \right) \approx \frac{3}{4}$. ♦

3B MORE ON APPLICATIONS OF PROBABILITY TO GENETICS

As we discussed in Companion for Chapter 1, inheritance patterns usually involve exactly two alleles per gene. However, there are genes that have multiple alleles. It is important to know that even when, in a population, multiple alleles of a gene exist, still, for that gene, each individual carries at most two different alleles. As an example, for blood groups in humans, there are three alleles: A, B, and O. Since each person inherits one of these alleles from each parent, there are six genotypes. Blood groups A and B are dominant to the O. Therefore, people who inherit type A from one parent and type O from another will have type A blood. Similarly, people who inherit type B blood from one parent and type O from another will have type B blood. Obviously, inheriting type A blood from both parents results in type A blood, inheriting type B blood from both parents results in type B blood, and inheriting type O blood from both parents results in type O blood. Interestingly enough, neither blood group A is dominant to blood group B nor vice versa. Individuals who inherit type A blood from one parent and type B blood from another have a blood type in which both A and B alleles are expressed. That blood type is called type AB blood. A and B are said to be **codominant** to each other.

In general, during fertilization, for each characteristic, each parent contributes *randomly* one member of its alleles *independently* of other characteristics and *independently* of the reproduction of other offspring. Therefore, in the pea plant example discussed previously, where A represents the dominant allele for purple flowers and a represents the recessive allele for white flowers, when the biologist crosses a pea plant of genotype AA with a pea plant of genotype aa , the resulting plant is Aa with probability 1. However, when crossing a plant of genotype Aa with another plant of genotype Aa , the resulting plant is AA with probability $1/4$, Aa with probability $1/2$, and aa with probability $1/4$. That is why, in the second generation, approximately $1/4$ th of the pea plants will have white flowers.

Example 3ℓ In guinea pigs, having long hair is a recessive trait. Suppose that an offspring of two short-haired guinea pigs has long hair. What is the probability that at least one of the next three offspring of these two guinea pigs also has long hair?

Solution: Let A represent the dominant allele for short hair and a represent the recessive allele for long hair. Since the offspring with long hair is aa , both parents must be Aa . Hence any other offspring is Aa with probability $1/2$, is AA with probability $1/4$, and is aa with probability $1/4$. Therefore, the probability is $3/4$ that any offspring is short-haired. The probability that the next three offspring are all short-haired is $(3/4)^3$. The probability that at least one of them is long-haired is $1 - (3/4)^3 = 37/64 = 0.578$.



Example 3m Consider the three alleles A, B, and O for blood types. Suppose that a man of genotype AB marries a woman of genotype AO. If they have six children, what is the probability that none will have type AB blood?

Solution: The genotype of a child of these parents will be one of the following with equal probabilities: AA, AB, AO, BO. Since the alleles A and B are codominant to each other and dominant to O, a child of these parents will have blood type A with probability $1/2$,

blood type AB with probability $1/4$, and blood type B with probability $1/4$. Therefore, by independence, the probability that none of the six children will have blood type AB is $(3/4)^6 \approx 0.178$. ♦

Example 3n Consider the three alleles A, B, and O for the human blood types. Suppose that, in a certain population, the frequencies of these alleles are 0.45, 0.20, and 0.35, respectively. Kim and John, two members of the population, are married and have a son named Dan. Kim and Dan both have blood types AB. John's blood type is B. What is the probability that John's genotype is BB?

Solution: Clearly, John's genotype is either BB or BO. Let E be the event that it is BB. Then E^c is the event that John's genotype is BO. Let F be the event that Dan's genotype is AB. By Bayes' formula, the probability we are interested in is

$$\begin{aligned} P(E | F) &= \frac{P(F | E)P(E)}{P(F | E)P(E) + P(F | E^c)P(E^c)} \\ &= \frac{(1/2)(0.20)^2}{(1/2)(0.20)^2 + (1/4)(0.20)(0.35)} = 0.53. \end{aligned}$$

Note that $P(E) = (0.20)^2$ and $P(E^c) = (0.20)(0.35)$ since for the two B alleles in John's blood type, each is coming from one parent *independently* of the other parent. ♦

Certain characteristics such as human height or human skin color result from **polygenic inheritance**. Unlike the color of the pea plant flowers, such characteristics are not controlled by a single gene. For example, it is the additive effect of several genes that determines the level of darkness of human skin. Apart from the environmental factors such as nutrition and light, in humans, variation in darkness level of the skin is continuous mainly because, as a heritable character, skin color is controlled by several genes. In humans, the same is true for the height and most other quantitative inheritance. Suppose that human skin color is controlled by four genes independently inherited. For those genes, let A, B, C , and D represent the dominant alleles for dark skin. For the respective genes, let a, b, c , and d represent the recessive alleles for light skin. Then individuals who are $AABBCCDD$ are the darkest members of the population, and individuals who are $aabbccdd$ are the lightest. Moreover, all the dominant alleles contribute to the skin color. For example, an individual who is $AAbbccdd$ is darker than an individual who is $Aabbccdd$, and those who are $Aabbccdd$ are darker than those with genotypes $aabbccdd$. Depending on the degree to which each of the four genes contributes to the darkness of the skin, there is a very wide range of skin colors among the members of the population. As an example, assuming that each gene contributes the same amount of darkness to the skin, people with the following genotypes are ordered from lightest to darkest:

$$Aabbccdd, \quad aaBbCcdd, \quad AABbccDd, \quad AaBbCcDD,$$

whereas people with the following genotypes are indistinguishable in skin color phenotype:

$$AAbbCcDd, \quad AaBbCcDd, \quad aaBbCcDD, \quad AABbCcdd.$$

As another example, suppose that on an island, the zeroth generation is the entire population, which consists of a male population all members having the lightest skin and a female population all members having the darkest skin. Suppose that a member belongs to the second generation if all of his or her grandparents belong to the zeroth generation. To find the probability that a member of the second generation has the lightest possible skin, of the four genes that control skin color, consider the gene whose dominant allele for dark skin is denoted by A . In the zeroth generation, all females are AA and all males are aa . Therefore, in the first generation, all members, regardless of gender, are Aa . In the second generation, a member is AA with probability $1/4$, Aa with probability $1/2$, and aa with probability $1/4$. Similarly, the probability is $1/4$ that in the second generation a member is bb , the probability is $1/4$ that he or she is cc , and the probability is $1/4$ that he or she is dd . Since genes are independently inherited, the probability that, in the second generation, a person has the lightest color (i.e., has genotype $aabbccdd$) is $(1/4)^4 = 1/256$.

Hardy-Weinberg Law

In organisms having two sets of chromosomes, called diploid organisms, which we all are, each hereditary character of each individual is carried by a pair of genes. A gene has alternate alleles that usually are dominant A or recessive a , so that the possible pairs of genes are AA , Aa (same as aA), and aa . Under certain conditions, by the Hardy-Weinberg law, to be discussed later, the probabilities of AA , Aa , and aa in all future generations remain the same as those in the second generation. To show this important fact, we will first discuss what is meant by *natural selection*, *mutation*, *random mating*, and *migration*.

Roughly speaking, **natural selection** is the process by which forms of life best adapted to the environment are the most fit. They live longer and are most likely to reproduce and hence perpetuate the survival of their type. Lack of natural selection implies that, in a form of life, all types are equally adapted to the environment. There is no type that survives longer and produces in greater number.

A *heritable* change to the genetic material is called **mutation**. It can happen by radiation, chemicals, or any carcinogen. A **mutagen** is a substance or preparation that can cause mutation.

In certain populations, individuals choose mates on the basis of their own genotypes. For example, among humans, short and tall individuals may marry more often within their own groups. In other situations, individuals might often choose to mate with individuals of different genotypes. In such cases, mating is not random. We say that, in a population, **random mating** occurs if mating between genotypes occur totally at random. Under random mating, between genotypes, mating will occur in proportion to the frequencies of the genotypes. Therefore, the probability of a mating between any two genotypes is the product of the frequencies of those genotypes in the population.

To prevent gene flow between populations of the same species, individuals should be prevented from **migration**. Clearly, immigrants and emigrants can change the balance of the frequencies of the genotypes. Existing alleles may be removed from the population by emigration. New alleles might be added by immigration. In both cases, the frequencies of the remaining alleles might increase or decrease.

Let r be the probability that a randomly selected person in the population is AA . Let s be the probability that he or she is Aa , and let t be the probability that he or she is aa . Then

$r + s + t = 1$. Let E be the event that a randomly selected father transmits A to a child. Then

$$\begin{aligned} P(E) &= P(E \mid \text{the father is } AA)P(\text{the father is } AA) \\ &\quad + P(E \mid \text{the father is } Aa)P(\text{the father is } Aa) \\ &\quad + P(E \mid \text{the father is } aa)P(\text{the father is } aa) \\ &= 1 \cdot r + \frac{1}{2} \cdot s + 0 \cdot t = r + \frac{s}{2}. \end{aligned}$$

Similarly, the probability that the father transmits a is $t + s/2$, the probability that the mother transmits A is $r + s/2$, and the probability that the mother transmits a is $t + s/2$. Now suppose that we have a very large population in which natural selection does not occur and is free from mutation and migration. Then, under random mating, the contributions of the parents to the genetic heritage of their offspring are independent of each other. Therefore, the probability that an offspring selected at random is AA is $r_1 = (r + s/2)^2$, the probability that he or she is Aa is $s_1 = 2(r + s/2)(t + s/2)$, and the probability that he or she is aa is $t_1 = (t + s/2)^2$. These formulas imply that, r_2 , the probability that a randomly selected offspring from the third generation is AA is given by $(r_1 + s_1/2)^2$, the probability that he or she is Aa is $s_2 = 2(r_1 + s_1/2)(t_1 + s_1/2)$, and the probability that he or she is aa is $t_2 = (t_1 + s_1/2)^2$. Some algebra shows that $r_2 = r_1$, $t_2 = t_1$ and $s_2 = s_1$. For example,

$$\begin{aligned} r_2 &= \left(r_1 + \frac{s_1}{2}\right)^2 = \left[\left(r + \frac{s}{2}\right)^2 + \left(r + \frac{s}{2}\right)\left(t + \frac{s}{2}\right)\right]^2 \\ &= \left[\left(r + \frac{s}{2}\right)\left(r + \frac{s}{2} + t + \frac{s}{2}\right)\right]^2 \\ &= \left[\left(r + \frac{s}{2}\right)(r + t + s)\right]^2 = \left(r + \frac{s}{2}\right)^2 = r_1. \end{aligned}$$

Therefore, the proportions of AA , Aa , and aa in the third generation are the same as in the second generation. Let $p = r + \frac{s}{2}$ and $q = t + \frac{s}{2}$. Then p is the frequency of allele A and q is the frequency of allele a in the population. We have shown that, under the conditions explained previously, when the population is in equilibrium, the genotype frequencies of AA , Aa , and aa are p^2 , $2pq$, and q^2 , respectively. Note that $p + q = 1$ implies that

$$p^2 + 2pq + q^2 = (p + q)^2 = 1,$$

as expected. We have established the following important law:

The Hardy-Weinberg Law: For a diploid organism with alternate dominant allele A and recessive allele a , the possible pairs of genes are AA , Aa , and aa . Under random mating conditions, in a very large population of a diploid organism in which natural selection does not occur and is free from mutation and migration, the proportions of AA , Aa , and aa in all future generations remain the same as those in the second generation. Moreover, if the gene frequency of A is p and the gene frequency of a is q , then, in all future generations, the genotype frequencies of AA , Aa , and aa remain in the proportions p^2 , $2pq$, and q^2 .

Example 3o Suppose that 64% of the sheep in a certain region have straight horn. If, for the sheep in that region, the allele for curly horn is dominant to the allele for straight horn, under the conditions of Hardy-Weinberg law, find the percentage of sheep who have two different alleles of the gene for horn shape.

Solution: Let the dominant allele for the curly horn be denoted by H and the recessive allele for straight horn be h . Since the genotype of a sheep with straight horn is hh , applying the Hardy-Weinberg's law, we have that $q^2 = 0.64$, or $q = 0.8$. Therefore, $p = 0.2$, and the percentage of sheep who have Hh is $2pq = 2(0.2)(0.8) = 0.32$. ♦

Sex-Linked Genes

In this subsection, we will discuss sex chromosomes and reproductive cells. Recall that the reproductive cells are the egg and sperm cells, collectively called **gametes**. We know that except for gametes, in each of the human cells, 23 chromosomes come from the father and 23 from the mother. Among the 23 pairs of chromosomes in each nongamete cell, there is one pair of chromosomes, called the **sex chromosomes**, which determines a person's gender. The pair of sex chromosomes in human females are called **X chromosomes**. Similar to other pairs of chromosomes, the pair of female sex chromosomes come one from each parent and look alike. The genes also appear in pairs, one from each parent, and those controlling the same characteristic occupy the same position on the paired X chromosomes inherited from father and mother. Concerning dominance and recessivity, the rules for the paired X chromosomes are the same as those followed by the 22 pairs of nonsex chromosomes of nongametes. The pair of sex chromosomes in human males do not look alike. They differ in shape and size. One of the chromosomes, inherited from the mother, is an X chromosome. The other one, inherited from the father, is said to be a **Y chromosome**. For most genes lying on the X chromosome, there is no corresponding gene on the Y chromosome. The only genes that lie on the Y chromosomes are those that are important in determining the male structures and organs. All genes on the X chromosome are called **X-linked genes**. Similarly, all genes on the Y chromosome are called **Y-linked genes**. For X -linked genes in males, we do not have recessivity and dominance. This is because males have exactly one X chromosome. Therefore, males carry only one allele of each X -linked gene. As a result, for example, if a male is color-blind, since the gene for color blindness is on the X chromosome, he must have inherited it from the mother. Some people bleed excessively when injured. They might have a genetic disease called hemophilia. If a male is a hemophiliac, he must have inherited it from his mother, since the genes for hemophilia lie on the X chromosome. There are many other characteristics such as X -linked baldness where a male inherits the characteristic from his mother.

Like humans, in all mammals, the sex chromosomes determine a mammal's gender. Mammalian males have one X chromosome and one Y chromosome. Mammalian females have two X chromosomes. In such cases, males are represented as XY , females as XX . It is important to know that the XY and XX patterns are not true for all males and females of all species. For example, female birds carry two different chromosomes, whereas male birds carry two identical chromosomes.

We will now discuss gametes. Unlike nonproductive cells, which have 23 pairs of chromosomes, gametes carry 23 single chromosomes. Therefore, no gamete contains any paired

chromosomes. A gamete carries every single gene of the individual, but in only one copy. When gametes are formed, with equal probabilities and independent of other genes, only one of the alleles of each gene will end up in the gamete. Clearly, a gamete has only one sex chromosome. In human females, the unique chromosome is necessarily an X chromosome. However, in males, it is either an X chromosome or a Y chromosome with equal probabilities. This follows because, in females, both of the sex chromosomes in nonreproductive cells are X and, in males, one of them is X and the other one is Y . Therefore, an egg contains only one X chromosome, while a sperm cell contains either an X chromosome or a Y chromosome with equal probability. If the sperm cell that fertilizes the egg carries an X chromosome, then the offspring is female. If it carries a Y chromosome, then the offspring is male. That explains why in humans it is equally likely that an offspring is a boy or a girl.

Example 3p What is the probability that 12 of the chromosomes in a randomly selected human gamete are paternal and 11 are maternal?

Solution: There is a one-to-one correspondence between the set of all possible combinations of 23 chromosomes of which 12 are paternal (P) and 11 are maternal (M) and the set of all distinguishable sequences of 12 P 's and 11 M 's. By the generalized counting principle, the number of sequences of P and M of length 23 is 2^{23} . By Theorem 2.4 of the book, the number of distinguishable permutations of 23 letters of which 12 are P and 11 are M is $\frac{23!}{12!11!}$. Therefore, the desired probability is $\left(\frac{23!}{12!11!}\right)/2^{23} = 0.16$. ♦

Example 3q Hemophilia is a sex-linked disease with normal allele H dominant to the mutant allele h . Kim and John are married and their son, Dan, has hemophilia. If the frequencies of H and h are 0.98 and 0.02, respectively, what is the probability that Kim has hemophilia?

Solution: Dan has inherited all of his sex-linked genes from his mother. Since Dan is hemophiliac, Kim is either Hh or hh . Let E be the event that she is hh . Then E^c is the event that she is Hh . Let F be the event that Dan has hemophilia. By Bayes' formula, the desired probability is

$$\begin{aligned} P(E | F) &= \frac{P(F | E)P(E)}{P(F | E)P(E) + P(F | E^c)P(E^c)} \\ &= \frac{1 \cdot (0.02)(0.02)}{1 \cdot (0.02)(0.02) + (1/2) \cdot [2(0.98)(0.02)]} = 0.02. \end{aligned}$$

Hence the conditional probability that Kim has hemophilia given that her son, Dan, has hemophilia is 0.02. Note that the unconditional probability that Kim has hemophilia is $(0.02)(0.02) = 0.0004$. ♦

EXERCISES FOR SECTION 3B

1. Kim has blood type O and John has blood type A. The blood type of their son, Dan, is O. What is the probability that John's genotype is AO?
2. Suppose that a gene has k ($k > 2$) alleles. For that gene, how many genotypes can an individual have?
3. For the shape of pea seed, the allele for round shape, denoted by R , is dominant to the allele for wrinkled shape, denoted by r . A pea plant with round seed shape is crossed with a pea plant with wrinkled seed shape. If half of the offspring have round seed shape and half have wrinkled seed shape, what were the genotypes of the parents?
4. In humans, the presence of freckles and having free earlobes are independently inherited dominant traits. Kim and John both have free earlobes and both have freckles, but their son, Dan, has attached earlobes and no freckles. What is the probability that Kim and John's next child has free earlobes and no freckles?
5. For *Drosophila* (a kind of fruit fly), B , the gray body, is dominant over b , the black body, and V , the wild-type wing is dominant over v , *vestigial* (a very small wing). A geneticist, T. H. Morgan, when mating *Drosophila* of genotype $BbVv$ with *Drosophila* of genotype $bbvv$, observed that 42% of the offspring were $BbVv$, 41% were $bbvv$, 9% were $Bbvv$, and 8% were $bbVv$. Based on these results, should we expect that the body color and wild-type genes of *Drosophila* are on different chromosomes (that is, are unlinked and hence independent)? Why or why not?
6. In a population, 1% of the people suffer from hereditary genetic deafness. The hearing allele is dominant and is denoted by D . The recessive allele for deafness is denoted by d . Kim and John, from the population, are married and have a son named Dan. If Kim and Dan are both deaf, what is the probability that John is also deaf?
7. In the United States, among the Caucasian population, cystic fibrosis is a serious genetic disease. For cystic fibrosis, the normal allele is dominant to the mutant allele. Suppose that, in a certain region of the country, 5.29% of the people have cystic fibrosis. Under the Hardy-Weinberg conditions, determine the percentage of people in that region who carry at least one mutant allele of the disease.
8. Hemophilia is a sex-linked disease with normal allele H dominant to the mutant allele h . Kim and John are both phenotypically normal. If the frequencies of H and h are 0.98 and 0.02, respectively, what is the probability that Dan, their son, has hemophilia?
9. Color blindness is a sex-linked hereditary disease with normal allele C dominant to the mutant allele c . Kim and John are married and their son, Dan, is color-blind. If, in the population, the frequencies of C and c are 0.83 and 0.17, respectively, what is the probability that Kim is color-blind?
10. Hemophilia is a sex-linked hereditary disease with normal allele H dominant to the mutant allele h . Kim and John are married and their daughter, Ann, has hemophilia.

If John has hemophilia, and the frequencies of H and h , in the population, are 0.98 and 0.02, respectively, what is the probability that Kim does not have hemophilia but she is a carrier for that disease?

11. In a certain country, all children with cystic fibrosis die before reaching adulthood. For this lethal disease, the normal allele, denoted by C , is dominant to the mutant allele, denoted by c . In that country, the probability is p that a person is only a carrier; that is, he or she is Cc . If Mr. J, a resident of the country had a brother who died of cystic fibrosis, what is the probability that Mr. J's next child will have cystic fibrosis?
12. Let p and q be positive numbers with $p + q = 1$. For a gene with dominant allele A and recessive allele a , let p^2 , $2pq$, and q^2 be the probabilities that a randomly selected person has genotype AA , Aa , and aa , respectively. If a man is of genotype AA , what is the probability that his brother is also of genotype AA ?
13. In human beings, the gene for naturally curly hair (H) is dominant over the gene for straight hair (h), and the gene for right-handedness (R) is dominant over the gene for left-handedness (r). Suppose that a left-handed man with curly hair marries a right-handed woman with straight hair. Find the probability that their child is (a) right-handed with curly hair; (b) right-handed with straight hair; (c) left-handed with curly hair; (d) left-handed with straight hair.
14. In human beings, brown eyes (B) are dominant over blue eyes (b). The probability that Stanley has genotype BB is $5/8$, and the probability that he has genotype bB is $3/8$. If Stanley marries Stella, a blue-eyed woman, what is the probability that their first child has brown eyes? Note that since all eye colors are dominant over blue, Stella's genotype is necessarily bb .
15. In genetics, a person or other organism is said to be a **carrier** of a certain trait if its genotype includes exactly one copy of a recessive allele of that trait. So a carrier of a trait does not display that trait. It is estimated that 3.2% of the US population is a carrier of the recessive hereditary gene of cystic fibrosis (c). Natasha is tested and is a carrier. However, we do not know whether her husband, Austin, is a carrier or not. We only know that he does not have cystic fibrosis. Find the probability that Austin is also a carrier if their first child does not have cystic fibrosis.
16. In human beings, the gene for naturally curly hair (H) is dominant over the gene for straight hair (h), and the gene for right-handedness (R) is dominant over the gene for left-handedness (r). Julie is a right-handed woman with curly hair. However, her genotype is not determined. The probability that her genotype is $HhRr$ is $2/3$, and the probability that it is $HhRR$ is $1/3$. If Julie marries a man with genotype $Hhrr$, what is the probability that their first child is right-handed with straight hair?



Answers to Odd-Numbered Exercises of Section 3B

- 1.** 1.
- 3.** The parents are rr and Rr .
- 5.** the observed data shows that the genes are linked.
- 7.** 40.71%.
- 9.** 0.17.
- 11.** $p/6$.
- 13.** The answers to parts (a), (b), (c), and (d) are $4/9$, $2/9$, $2/9$, and $1/9$, respectively.
- 15.** 0.024.